

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 15)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain what a carbon footprint should include when calculated for a car.

[3 marks]

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0 1 . 2

Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

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0 1 . 3

Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

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0 2 . 1

Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

0 2 . 2

Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

0 2 . 3

Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

5

0 3

Table 1 shows the number of plant species per m^2 in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m^2	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

03 . 1

Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

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03 . 2

Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

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03 . 3

Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

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03 . 4

Explain how burning maintains heathland.

[2 marks]

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04

Table 1 shows the number of bird species and butterfly species recorded in three woodlands in the same region.

Table 1

Woodland	Management	Bird species	Butterfly species
A	Ancient oak wood, coppiced on rotation with clearings	42	23
B	Ancient oak wood, unmanaged for 60 years, dense canopy	28	9
C	40-year-old conifer plantation	14	3

04

. 1

Use Table 1 and your knowledge to explain the differences in biodiversity between the three woodlands.

[5 marks]

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04

. 2

Describe **three** conservation efforts used to protect UK woodlands.

[3 marks]

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3

0 4 . 3 Explain why a high biodiversity increases the ecological stability of a woodland. **[2 marks]**

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0 5 . 1 Explain how each of the following agricultural changes affects wildlife habitats. **[6 marks]**

Drainage of wet grassland

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Application of fertiliser to a species-rich meadow

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Removal of hedgerows to enlarge fields

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0 5 . 2 Explain why some farmland habitats are valuable for wildlife conservation. **[2 marks]**

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0 5 . 3 Describe how agriculture contributes to climate change. **[2 marks]**

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0 6 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 6 . 1 Complete Table 1 by adding the missing adaptation or application. **[4 marks]**

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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06 . 2

Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

06 . 3

Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

1

2

06 . 4

Explain why bird wing bones are a useful model for engineers.

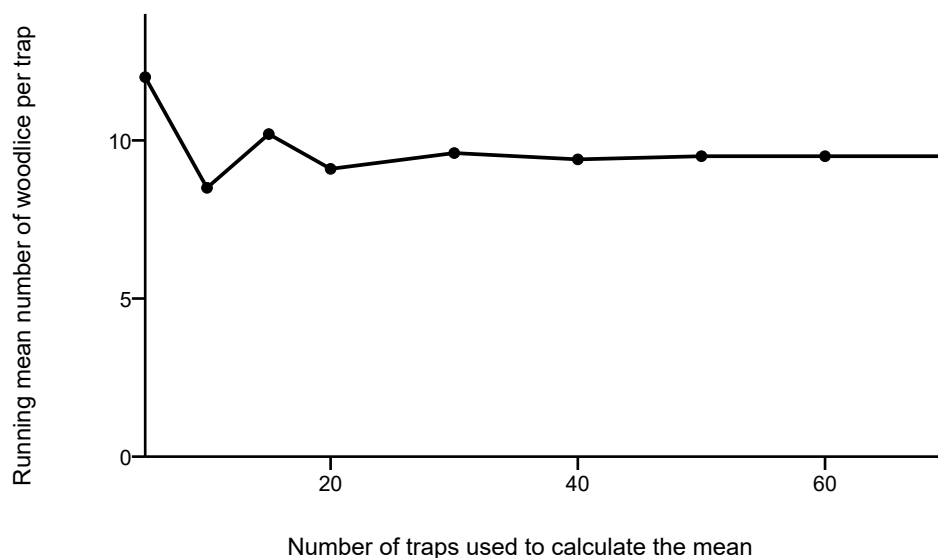
[1 mark]

10

07

Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1

**07****1**

Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

07**2**

Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

07**3**

The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

07 . 4

Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

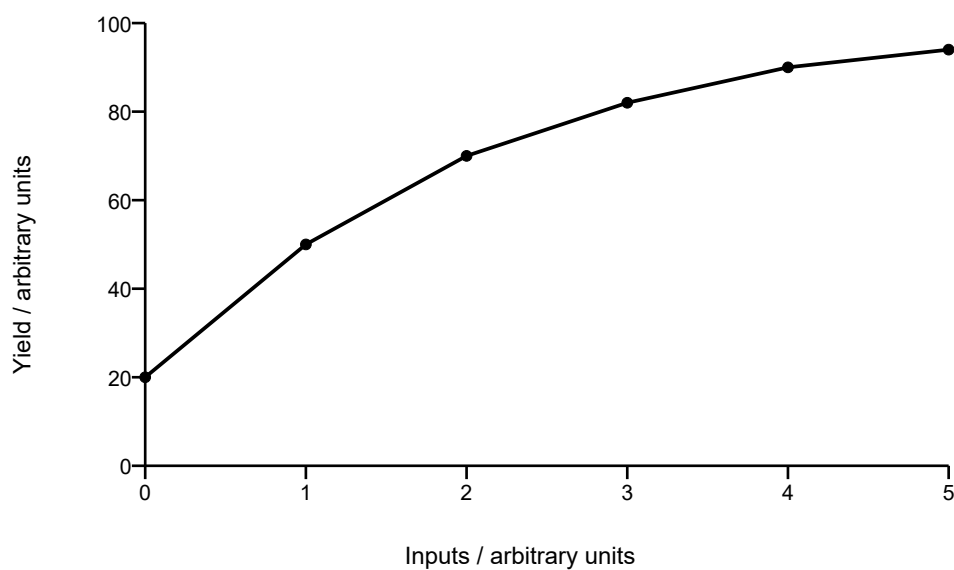
[3 marks]

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08

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



08 . 1

Describe the relationship shown in Figure 1.

[2 marks]

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08 . 2

Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer

units

08 . 3

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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08 . 4

Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

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08 . 5

Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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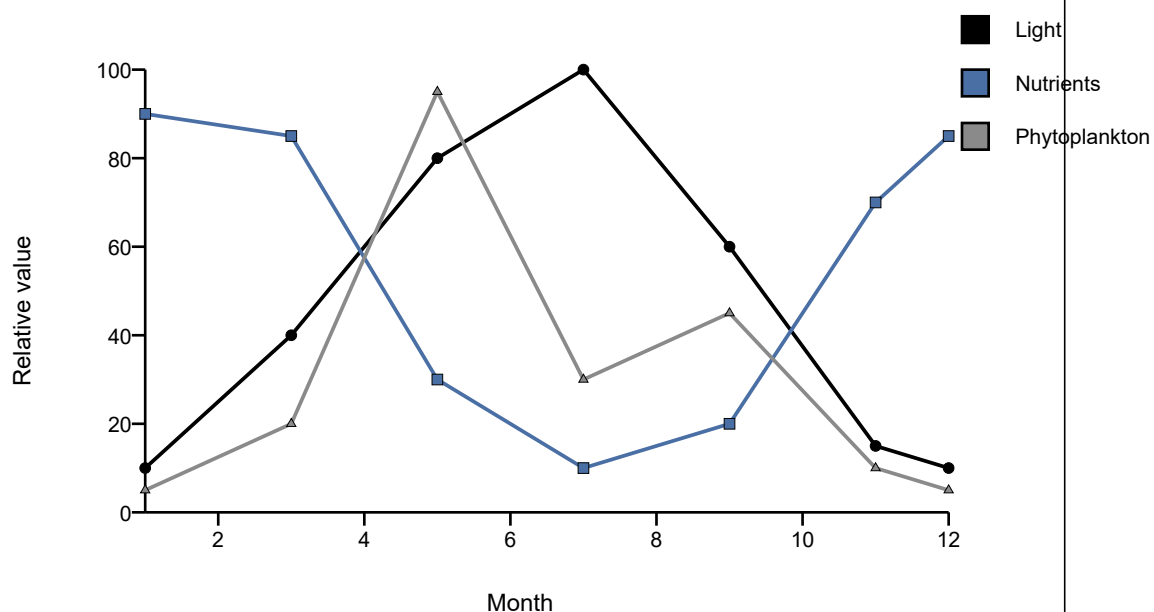
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09

Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



09

1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

09

2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

09 . 3

Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

09 . 4

Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

10

10 . 1

Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

[illegible]

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10 . **2** State what is meant by a **fed** aquaculture species. **[1 mark]**

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Do not write outside the box

Write an essay on one of the following topics.

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Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

OR

1 1 . 2

Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS